

PARUL UNIVERSITY**FACULTY OF IT & COMPUTER SCIENCE****BCA / BScIT/BCA(HONS)/BScIT(HONS)/IMC/MCA/M.Sc.IT 2025-26 Mid-Term Examination****Semester: 3****Date: - -2025****Subject Code: 05201349****Time: (1hr:30min)****Subject Name: Full Stack Web Development – II****Total Marks: 40****Instructions:**

- Figures to the right indicate full marks.
- Make suitable assumptions wherever necessary.

Q.1	Answer the following.	CO	PO	BL	[10]
(a)	3 short questions of 1 mark each				[3]
	(i) What is the role of the Socket class in Java networking?	1	1	Remembering (BT-1)	
	(ii) Define RMI.	1	1	Remembering (BT-1)	
	(iii) What is JavaFX primarily used for in Java?	3	1	Remembering (BT-1)	
(b)	Objective type/MCQs/True-False/Fill in blanks (7 questions of 1 mark each)				[7]
	1. Which class is used to get the IP address of a machine in Java? a) InetAddress b) Socket c) ServerSocket d) DatagramSocket	1	1	Remembering (BT-1)	
	2. True or False: Java RMI allows an object running in one JVM to invoke methods on an object in another JVM. Justify your answer	1	2	Understanding (BT-2)	
	3. The _____ class in Java can be used to read from a URL.	1	1	Remembering (BT-1)	
	4. Which of the following is NOT a 2D shape in JavaFX? a) Rectangle b) Sphere c) Circle d) Ellipse	3	1	Remembering (BT-1)	
	5. Which method is called to launch a JavaFX application?	3	1	Understanding (BT-2)	
	6. What is the purpose of stub and skeleton in RMI?	1	2	Remembering (BT-1)	
	7. JavaFX supports the creation of animation effects. (True/False)	3	1	Understanding (BT-2)	
Q.2	Answer the following. (2 or 3 mark questions)				[10]
(a)	Two Questions of 2 Marks				[4]
	(i) Explain how a Java application establishes a connection to a server using sockets.	1	2	Understanding (BT-2)	(02)
	(ii) Briefly describe the architecture of JavaFX.	3	1,3	Understanding (BT-2)	(02)

					[6]
(b)	Two Questions of 3 Marks			Apply (BT-3)	(03)
	(i) Write the steps involved in developing and running a simple Java RMI application.	1	1,5		
	(ii) How do you implement a basic 2D shape and event handling in JavaFX? Provide a short code example.	3	3,5	Apply & Analyse (BT-3 & 4)	(03)
Q.3	Attempt any TWO.				[10]
	(i) Explain the differences between TCP and UDP sockets in Java. Provide use cases for each.	1	1	Apply (BT-3)	(05)
	(ii) Design a scenario where you would use Java RMI to enable client-server communication. Explain the working with a diagram.	1	3,5	Analyse (BT-4)	(05)
	(iii) "Describe the animation capabilities in JavaFX and how they can be applied in an interactive desktop application." Illustrate with an example.	3	3,5	Evaluate (BT-5)	(05)
Q.4	Answer the following.				[10]
(a)	A company needs to build a client-server application where multiple clients can request current data from a central server. Draft the architecture using Java networking classes, and describe how data will be sent and received.	1	2,3,10	Analyze & Evaluate (BT-4 & 5)	(05)
(b)	Develop a simple JavaFX application that displays two geometric shapes. Add event handlers to change the color of the shapes when clicked. Explain the event model used.	3	3,5	Analyse (BT -3)	(05)
OR					
(b)	Modify a JavaFX application so that when a user clicks a button, the application fetches and displays the current date and time from a server using Java networking concepts (e.g., sockets). Explain how your application manages potential network errors.	3	1,2,5	Evaluate (BT-5)	(05)